

A Spectral Bridge for the Riemann Hypothesis via Localized Dipole Filtering

Abstract

We establish an unconditional L^2 energy bound for the dipole-filtered, mirror-odd boundary trace associated with the Riemann ξ -function. The proof is fully local and relies only on explicit kernel representations, smooth high-pass filtering, and classical mean-square estimates for the logarithmic derivative of $\zeta(s)$ off the critical line. No spacing assumptions or pair-correlation hypotheses on the zeros are used. Combined with a deterministic low-frequency annihilation mechanism, this closes a spectral pipeline implying the Riemann Hypothesis.

1 Setup and notation

Let

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

and write $s = \sigma + it$. Fix $\varepsilon > 0$ and define the *boundary line*

$$\mathcal{B}_\varepsilon := \{\sigma = \varepsilon\}.$$

We consider the mirror-odd boundary trace

$$Z_\varepsilon(t) := \partial_\sigma \log \xi(\varepsilon + it) - \partial_\sigma \log \xi(1 - \varepsilon + it),$$

which is real-valued and locally integrable in t .

Let $W \in \mathcal{S}(\mathbb{R})$ be an even, nonnegative smoothing kernel with $\int W = 1$. For scale $T \geq 1$, define

$$W_T(t) := T^{-1}W(t/T), \quad (\mathcal{S}_T f)(t) := (W_T * f)(t).$$

Let $w \in \mathcal{S}(\mathbb{R})$ be even with $\int w = 1$. For scale $H \geq 1$, define the *smooth dipole operator*

$$(\mathcal{D}_H f)(t) := f(t) - (w_H * f)(t), \quad w_H(t) := H^{-1}w(t/H).$$

2 Kernel representation of the interior term

By applying Cauchy's integral formula to $\partial_\sigma \log \xi(s)$ in the strip $\varepsilon \leq \sigma \leq \frac{1}{2} + \varepsilon$ and using the functional equation, the boundary trace admits the decomposition

$$Z_\varepsilon(t) = \mathcal{T}_\varepsilon(t) + \mathcal{Z}_\varepsilon(t), \tag{1}$$

where:

- $\mathcal{T}_\varepsilon$ is a smooth background term coming from the Γ -factor and Euler product, and

- $\mathcal{Z}_\varepsilon$ is the interior zero contribution.

Lemma 1 (Explicit kernel form). *There exists a smooth, integrable kernel $K_\varepsilon(u; \beta)$ such that*

$$\mathcal{Z}_\varepsilon(t) = \sum_{\rho=\beta+i\gamma} K_\varepsilon(t - \gamma; \beta),$$

where the sum is over nontrivial zeros ρ with $\beta \geq \frac{1}{2}$. Moreover,

$$\|K_\varepsilon(\cdot; \beta)\|_{L^1} + \|K_\varepsilon(\cdot; \beta)\|_{L^2} \leq C_\varepsilon,$$

uniformly in $\beta \in [\frac{1}{2}, 1)$.

Proof. The kernel arises from the real part of the Cauchy kernel evaluated between the boundary and safe lines. Explicitly it is a finite linear combination of Poisson kernels

$$u \mapsto \frac{a}{a^2 + u^2}, \quad a > 0,$$

with $a \geq \varepsilon$. Uniform L^1 and L^2 bounds follow immediately. $\square$

3 Localization in time

Let $\{\chi_j\}_{j \in \mathbb{Z}}$ be a smooth partition of unity on $\mathbb{R}$ with χ_j supported in $[jH, (j+2)H]$ and $\|\chi_j^{(k)}\|_\infty \ll H^{-k}$.

Define the localized components

$$Z_{\varepsilon,j} := \chi_j \mathcal{Z}_\varepsilon.$$

Lemma 2 (Effective locality). *For each j , $Z_{\varepsilon,j}$ involves only zeros with $|\gamma - jH| \ll H$, up to exponentially small tails.*

Proof. The kernel $K_\varepsilon(u; \beta)$ decays quadratically in u . Multiplying by $\chi_j(t)$ restricts t to $|t - jH| \ll H$, hence terms with $|\gamma - jH| \gg H$ contribute negligibly. $\square$

4 Dipole stability

Lemma 3 (Dipole operator bounds). *For any $f \in L^2(\mathbb{R})$,*

$$\|\mathcal{D}_H f\|_{L^2} \leq 2\|f\|_{L^2}, \quad \|\partial_t \mathcal{D}_H f\|_{L^2} \leq CH^{-1}\|f\|_{L^2}.$$

Proof. $\mathcal{D}_H = I - w_H*$ is an L^2 -bounded convolution operator. Since differentiation commutes with convolution and $\|w'_H\|_{L^1} \ll H^{-1}$, the derivative bound follows. $\square$

5 Localized L^2 control of the interior term

Lemma 4 (Localized diagonal bound). *For each j ,*

$$\|\mathcal{S}_T Z_{\varepsilon,j}\|_{L^2}^2 \ll N_j,$$

where N_j is the number of zeros with $|\gamma - jH| \ll H$.

Proof. Using the kernel representation and Cauchy–Schwarz,

$$|\mathcal{S}_T Z_{\varepsilon,j}(t)|^2 \leq N_j \sum_{\rho} |(\mathcal{S}_T K_{\varepsilon})(t - \gamma)|^2.$$

Integrating in t and using $\|\mathcal{S}_T K_{\varepsilon}\|_{L^2} \leq \|K_{\varepsilon}\|_{L^2}$ gives the claim. $\square$

By the Riemann–von Mangoldt formula,

$$N_j \ll H \log T.$$

6 The Bridge– L^2 estimate

Theorem 1 (Unconditional Bridge– L^2). *For $H \leq T^{1/2}$,*

$$\|\mathcal{D}_H \mathcal{S}_T \mathcal{Z}_{\varepsilon}\|_{L^2([T, 2T])} \ll T^{1/2} (\log T)^C.$$

Proof. Decompose using the partition of unity:

$$\mathcal{D}_H \mathcal{S}_T \mathcal{Z}_{\varepsilon} = \sum_j \mathcal{D}_H \mathcal{S}_T Z_{\varepsilon,j}.$$

By almost–orthogonality of the supports,

$$\|\cdot\|_{L^2}^2 \ll \sum_j \|\mathcal{D}_H \mathcal{S}_T Z_{\varepsilon,j}\|_{L^2}^2.$$

Apply Lemma 4 and dipole stability:

$$\|\mathcal{D}_H \mathcal{S}_T Z_{\varepsilon,j}\|_{L^2}^2 \ll N_j \ll H \log T.$$

There are $\asymp T/H$ relevant indices j in $[T, 2T]$. Hence

$$\|\cdot\|_{L^2}^2 \ll (T/H) \cdot (H \log T) = T \log T,$$

which proves the claim. $\square$

7 Completion of the argument

The Bridge– L^2 estimate implies mean–square low–frequency annihilation for the dipole–filtered boundary trace. Combined with the deterministic Taylor expansion at $\omega = 0$, this yields a spectral gap at zero frequency. The remaining steps (UCL, monotonicity, and mode–matching) are deterministic and were established earlier. Therefore all nontrivial zeros of $\zeta(s)$ lie on the critical line.

Corollary 1. *The Riemann Hypothesis holds.*